PENGZHEN LI

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SUMMARY

Final-year Economics Ph.D. candidate (expected May 2027) specializing in applied econometrics, causal inference, and optimization. Four peer-reviewed publications (one with 950+ citations), five working papers including a job market paper on trade restrictiveness with James Lake, 2024 Holly Award. Build reproducible analytics in Stata/R/Python/SQL/GAMS and turn large administrative, trade, and production datasets into decisions. Authorized to work in the U.S. without sponsorship.

EDUCATION

University of Tennessee, Knoxville — Jan 2020 – May 2027 (expected)
Ph.D. in Economics — fields: international trade, applied econometrics, environmental and resource economics

Shandong University — Sep 2016 – Jun 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (950+ citations)

Tongji University, Shanghai — Sep 2011 – Jun 2015
B.A. in Economics

SKILLS

Programming / software: Python (pandas, numpy, statsmodels, scikit-learn), R, Stata, SQL, GAMS, MATLAB, ArcGIS, IMPLAN, Git, LaTeX.
Methods: causal inference (DiD, event study, IV, RDD, synthetic control, fixed effects), panel and time-series econometrics (GMM, panel threshold, TVAR, VECM, mGARCH), forecasting, mixed-integer and stochastic optimization, DEA, game theory (Shapley value, mean-field games), supervised and unsupervised machine learning, text-as-data.
Credentials / languages: CFA Level II; English (fluent), Mandarin (native).

RESEARCH EXPERIENCE

Doctoral Researcher, Department of Economics, University of Tennessee — 2023 – present

- *Trade restrictiveness under discriminatory tariffs* (with J. Lake; job market paper): derived welfare-based indices for partner-specific tariffs and a level/discrimination decomposition; built the WITS/Comtrade pipeline for 2017–2024 and quantified how MFN-only indices understate trade-war distortions. Presented at the Midwest Trade Meetings, 2025.
- *Multilateral tariff bargaining* (with G. Schaur): cooperative-game (Shapley value) model of coalition stability and gains from liberalization, calibrated to WTO bilateral flows for RCEP/CPTPP scenario analysis.
- *Mayoral endorsements and presidential elections* (with M. Van Essen, L. Lima): mean-field-game model of strategic endorsement timing tested on Brazilian and French administrative data; 2024 Holly Award.

Graduate Research Assistant, University of Tennessee — Jan 2020 – Aug 2023
USDA NIFA and FAA ASCENT funded projects on sustainable-aviation-fuel supply chains.

- Built mixed-integer optimization models (GAMS) to design cost- and profit-optimal forest-residue sustainable-aviation-fuel supply chains across the Southeast U.S.; published in *Biomass and Bioenergy*

(2026) and *Transportation Research Record* (2024).

- Ran facility-siting and regional impact analysis with IMPLAN and efficiency analysis with DEA; delivered scenario and sensitivity results (EMP, supervised ML, @RISK) to an interdisciplinary engineering–economics team and federal sponsors.
- Engineered reproducible ArcGIS/R/Python pipelines for high-resolution spatial data.

Earlier applied projects 2016 – 2020

- Panel-threshold evidence on green innovation and CO_2 (*TFSC* 2019, 950+ citations); synthetic-control estimate of the Wenchuan earthquake's long-run output effect; GMM analysis of MFP/CFAP payments and U.S. agricultural lending; TVAR/VECM/mGARCH commodity-price models; equity-premium prediction from *NYT*/*WSJ* text.

SELECTED PUBLICATIONS

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." With T. Yu et al. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." With T. Yu et al. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." With C. Martinez et al. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." With K. Du and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

TEACHING

Instructor: ECON 351 Monetary Economics (Spring 2025, 2026); ECON 322 The Global Economy: Trade and Development (Summer 2025); ECON 313 Intermediate Macroeconomics (Summer 2026).
Teaching assistant: Ph.D. Microeconomic Theory (ECON 511/512), Mathematical Methods (ECON 581), Agribusiness Operations Research (AREC 525).

HONORS

J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee 2024